

SAT Hard Question 4

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Question 1: Probability (Statistics)

The incomplete table below summarizes the number of left-handed and right-handed students by gender for eighth-grade students at Keisel Middle School.

Gender	Left	Right
Female		
Male		
Total	18	122

There are **5 times** as many right-handed female students as left-handed female students, and **9 times** as many right-handed male students as left-handed male students.

If there are a total of 18 left-handed and 122 right-handed students, which of the following is closest to the probability that a randomly selected right-handed student is female?

- A. 0.410
- B. 0.357
- C. 0.333
- D. 0.250

Answer to Question 1

Let x be the number of left-handed females and y be the number of left-handed males.

- Total left-handed: $x + y = 18$
- Right-handed females: $5x$, right-handed males: $9y$
- Total right-handed: $5x + 9y = 122$

Now solve the system:

$$x + y = 18$$

$$5x + 9y = 122$$

Substitute $y = 18 - x$ into the second equation:

$$\begin{aligned} 5x + 9(18 - x) &= 122 \Rightarrow 5x + 162 - 9x = 122 \Rightarrow -4x = -40 \Rightarrow x = 10 \\ \Rightarrow y &= 8, \quad \text{Right-handed females} = 50, \quad \text{Right-handed students} = 122 \end{aligned}$$

$$\text{Probability} = \frac{50}{122} \approx 0.41$$

Correct Answer: A. 0.410

Question 2: Little's Law

If shoppers enter a store at an average rate of r shoppers per minute and each stays in the store for an average of T minutes, the average number of shoppers in the store is $N = rT$.

In the original store:

$$r = 3 \text{ shoppers/minute}, \quad T = 15 \Rightarrow N = 45$$

In the new store:

$$r = \frac{90}{60} = 1.5, \quad T = 12 \Rightarrow N = 1.5 \times 12 = 18$$

What percent less is the new store's average number of shoppers compared to the original?

- A. 57
- B. 58
- C. 59
- D. 60

Answer to Question 2

From Little's Law:

$$\text{Original store: } N = 3 \times 15 = 45$$

$$\text{New store: } r = \frac{90}{60} = 1.5, \quad T = 12, \quad N = 1.5 \times 12 = 18$$

$$\text{Percent decrease} = \frac{45 - 18}{45} \times 100 = \frac{27}{45} \times 100 = 60\%$$

Correct Answer: D. 60

Question 3: Statistical Generalization

A trivia tournament organizer wanted to study the relationship between the number of points a team scores in a trivia round and the number of hours that a team practices each week.

For the study, the organizer selected **55** teams at random from all trivia teams in a certain tournament. The table displays the information for the **40** teams in the sample that practiced for at least **3 hours per week**.

Hours practiced	6 to 13 points	14 or more points	Total
3 to 5 hours	6	4	10
More than 5 hours	4	26	30
Total	10	30	40

Which of the following is the **largest population** to which the results of the study can be generalized?

- A. All trivia teams in the tournament that scored **14 or more points**.
- B. The **55 trivia teams** in the sample.
- C. The **40 trivia teams** in the sample that practiced for at least **3 hours per week**.
- D. All trivia teams in the **tournament**.

Answer to Question 3

Correct Answer: D.

Question 4: Tangent Line and Circle Geometry

Given: The circle has center $(-1, 1)$. Line t is tangent to this circle at point $(4, -3)$.

Which of the following points lies on line t ?

- A. $(0, \frac{5}{4})$
- B. $(3, 6)$
- C. $(8, 2)$
- D. $(9, 1)$

Answer to Question 4

To find the tangent line to a circle at a point, find the radius vector from the center to the point of tangency:

$$\vec{r} = \langle 4 - (-1), -3 - 1 \rangle = \langle 5, -4 \rangle$$

The slope of the radius is $\frac{-4}{5}$, so the slope of the tangent line is the negative reciprocal: $\frac{5}{4}$.

Using point-slope form:

$$y + 3 = \frac{5}{4}(x - 4)$$

Now test which point lies on this line. Try (8, 2):

$$2 + 3 = \frac{5}{4}(8 - 4) \Rightarrow 5 = \frac{5}{4} \cdot 4 = 5$$

Correct Answer: C. (8, 2)

Question 5: Rectangular Prism Surface Area

Two identical rectangular prisms each have a height of **90 cm**. The base of each prism is a square, and the surface area of each prism is $K \text{ cm}^2$. When the prisms are glued together along one square base, the new surface area is:

$$\frac{92}{47}K \text{ cm}^2.$$

What is the side length, in cm, of each square base?

- A. $s = 4$
- B. $s = 6$
- C. $s = 8$
- D. $s = 10$

Answer to Question 5

Let the side length be s . For a rectangular prism with square base:

$$\text{Surface Area} = 2s^2 + 4sh$$

Each original prism has:

$$K = 2s^2 + 4s(90) = 2s^2 + 360s$$

When glued together along one square base, one face is removed:

$$\text{New surface area} = 2K - 2s^2 = 2(2s^2 + 360s) - 2s^2 = 2s^2 + 720s$$

Given:

$$2s^2 + 720s = \frac{92}{47}(2s^2 + 360s)$$

Solve for s , and you'll find $s = 8$

Correct Answer: C. $s = 8$

Question 6: Triangle Similarity and Area

In triangle RST , angle T is a right angle. Point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$.

Given:

- $RT = 72$ units,
- $LK = 24$ units,
- Area of $\triangle RST = 792$ square units.

Find the length of KT , in units.

Answer to Question 6

Since $LK \parallel RT$, triangle $\triangle LKT \sim \triangle RST$ by AA similarity.

$$\frac{LK}{RT} = \frac{24}{72} = \frac{1}{3}$$

Because area scales by the square of the side ratio:

$$\text{Area of } \triangle LKT = \left(\frac{1}{3}\right)^2 \cdot 792 = \frac{1}{9} \cdot 792 = 88$$

Now use triangle area formula:

$$\text{Area} = \frac{1}{2} \cdot LK \cdot KT = 88 \Rightarrow \frac{1}{2} \cdot 24 \cdot KT = 88 \Rightarrow 12 \cdot KT = 88 \Rightarrow KT = \frac{88}{12} = \frac{22}{3}$$

Correct Answer: $\frac{22}{3}$ units

Question 7: Circle Equation (Geometry)

Given the circle equation:

$$(x - 6)^2 + (y + 5)^2 = 16$$

Point $P(10, -5)$ lies on the circle, and segment PQ is the diameter.

- What are the coordinates of point Q ?

Answer to Question 7

- The center of the circle is the midpoint of diameter PQ , and is given as:

$$(6, -5)$$

- Let point $Q = (x, y)$, and recall that the midpoint of PQ is:

$$\left(\frac{10 + x}{2}, \frac{-5 + y}{2} \right) = (6, -5)$$

- Solve for x and y :

$$\frac{10 + x}{2} = 6 \Rightarrow 10 + x = 12 \Rightarrow x = 2$$

$$\frac{-5 + y}{2} = -5 \Rightarrow -5 + y = -10 \Rightarrow y = -5$$

Coordinates of point Q : $(2, -5)$